ZS-M15
Falsification-Based and McKay-Structural Upgrade of ZS-M9 Table 2: From HYPOTHESIS Strong to DERIVED

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March 2026 — ZS-M15 (Mathematical Spine Theme)

Verification: 21/21 PASS | Zero Free Parameters

§0. Abstract

We upgrade the ZS-M9 Table 2 assignment of icosahedral rotation group $I \cong A_5$ irreducible representations to Standard Model field classes from HYPOTHESIS strong (five lines of evidence) to DERIVED. Two complementary reductions combine to yield this upgrade. Route (b) Exhaustive Falsification reduces the 120-element permutation space to exactly two assignments using only two PROVEN inputs from ZS-M9: Theorem 3.1 (chirality classification) and Theorem 3.2 (gauge dimension saturation). The remaining binary ambiguity ($3 \leftrightarrow 3'$ swap under LH/RH labels) is then broken by the Z5-McKay Handedness Theorem (this paper, DERIVED), which invokes the ZS-M9 Table 4 McKay bridge $Z_5 \to \hat{A}_4 \to SU(5)$ (DERIVED) together with the PROVEN Z_5 charge complementarity of §4 F1: I-irrep 3 carries the SU(2)_L simple root ω^4 while I-irrep $3'$ carries the U(1)_Y simple root ω^3, forcing the left-handed assignment to I-3 under SM handedness. A 500,000-sample three-basket Monte Carlo confirms that the $120 \to 1$ reduction is structural rather than coincidental: random chirality-plus-saturation frameworks produce a unique selection at $p < 0.0001\%$, and the ordered dimension partition (1, 3, 3, 4, 5) is the unique admissible solution to $\Sigma d_i^2 = 60$ (OBSERVATION). We document three downstream upgrades: ZS-S11 §3.1 Witten anomaly, ZS-S11 §3.2 A1 color anomaly, and ZS-M11 §9.5.3 m_{D,1} = 0 Dirac mass, each of which cascades from DERIVED-CONDITIONAL to DERIVED once ZS-M15 is accepted. The full PROVEN upgrade, requiring an action-level dynamical selection mechanism (Route (a) of ZS-U9 §8.2), remains open and is transferred to future work. Zero new free parameters beyond A = 35/437; the 21-test verification suite passes at machine precision.

Keywords: icosahedral symmetry, A_5 representation theory, McKay correspondence, Standard Model field assignment, Gap G2 closure, exhaustive falsification, chirality classification, gauge dimension saturation, Z5 charge complementarity, anti-numerology.

Epistemic Status Legend

STATUS	DEFINITION

PROVEN	Mathematical theorem with complete proof under declared definitions. Falsifiable only by logical or computational error.
DERIVED	Rigorous argument using one or more DERIVED or PROVEN ingredients from upstream papers. Physical or representation-theoretic derivation.
DERIVED-CONDITIONAL	DERIVED status contingent on one or more upstream HYPOTHESIS-level results; upgrades automatically when the upstream result is upgraded.
HYPOTHESIS (strong)	Proposed identification supported by multiple independent lines of evidence but lacking a single decisive derivation.
OBSERVATION	Numerical or structural fact recorded without derivation; flagged for future structural explanation if available.
OPEN	Identified gap or subcomputation pending; scope of consequence documented.
NON-CLAIM	Explicit declaration of what this paper does NOT establish, to prevent overclaim.
LOCKED	Input value fixed from prior paper; not adjustable within this paper.

§1. Introduction

§1.1 The Gap G2 Problem

The Z-Spin Trinity Braiding Theorem (ZS-U9 v1.0, DERIVED) [1] produces the complete Standard Model hypercharge spectrum $Y_Q = +1/6$, $Y_u = +2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$, $Y_{\nu_R} = 0$ from four ingredients combining compact phase quantization (Ingredient I), Yukawa uniqueness (Ingredient II, ZS-M10 Theorem 2.1 PROVEN) [2], the McKay bridge SU(5) Cartan (Ingredient III, ZS-M9 §5 DERIVED) [3], and neutral-Higgs hypercharge fixing (Ingredient IV / Theorem T3 DERIVED, 2026-04-19 update to ZS-U9). One upstream gap, documented explicitly in ZS-U9 §8.2 as Gap G2, prevents the Trinity Braiding from reaching full PROVEN status: the assignment of $I \cong A_5$ irreducible representations to Standard Model field classes given by ZS-M9 Table 2 carries the epistemic tag HYPOTHESIS strong.

ZS-U9 §8.2 records the assignment as:

$$\nu_R \leftrightarrow 1, \quad \text{LH fermions} \leftrightarrow 3, \quad \text{RH fermions} \leftrightarrow 3', \quad \text{gauge bosons} \leftrightarrow 4, \quad \text{Higgs} \leftrightarrow 5,$$

and documents five independent lines of supporting evidence: (i) chirality index Δ (ZS-M9 Theorem 3.1 PROVEN); (ii) A_4 content under the $A_4 \subset I$ subgroup embedding; (iii) D_5 content under the pentagonal stabilizer $D_5 \subset I$ subgroup embedding; (iv) gauge dimension saturation (ZS-M9 Theorem 3.2 PROVEN); (v) complete branching rules across the six physically relevant subgroups of I. ZS-U9 §8.2 further identifies two complementary resolution paths: Route (a) derivation from a Z-Spin-native potential minimization on the space of I-equivariant embeddings, analogous to the ZS-S4 §6.12 Higgs VEV selection of the Φ-attractor; and Route (b) proof of uniqueness by exhaustive falsification of alternative

assignments. Route (b) is noted as partially complete via Theorem 3.2 PROVEN, singling out {3, 3′} as the unique pair of fermion irreps saturating dim(ρ ⊗ 4) = G = 12.

§1.2 What This Paper Does and Does Not Do

This paper delivers Route (b) in its complete form and adds a new Z5-McKay Handedness Theorem that breaks the residual 3 ↔ 3′ ambiguity. The combination reaches DERIVED status for Table 2.

This paper IS: (i) an explicit enumeration over the full 120-element permutation space of I-irrep-to-SM-field-class maps, showing that only two assignments survive the combined chirality and gauge-dimension-saturation constraints of ZS-M9 Theorems 3.1 and 3.2; (ii) a new Z5-McKay Handedness Theorem that uses the ZS-M9 Table 4 McKay bridge (DERIVED) to identify the SU(2)_L simple root ω^4 exclusively with I-irrep 3, breaking the residual 3 ↔ 3′ ambiguity in favor of the Table 2 assignment; (iii) a three-basket 500,000-sample anti-numerology Monte Carlo confirming structural selectivity; (iv) a formal status upgrade declaration from HYPOTHESIS strong to DERIVED, with documented downstream cascade upgrades in ZS-S11, ZS-M11, and ZS-U9.

This paper IS NOT: (i) a closure of Gap G2 at the PROVEN level — PROVEN requires Route (a) action-level dynamical selection, which is deferred to future work; (ii) an independent rederivation of the electron subspace identification in ZS-M14 — while ZS-M14 Corollary III (DERIVED) is consistent with Assignment A, it inherits the Table 2 HYPOTHESIS strong assignment upstream via ZS-S9 Pillar I and therefore does not provide an independent closure of Gap G2; (iii) a derivation of new numerical predictions — all numerical content is inherited from upstream papers with the status of Table 2 upgraded without altering any numerical result; (iv) an introduction of new free parameters — A = 35/437, Q = 11, (Z, X, Y) = (2, 3, 6) remain the sole Z-Spin geometric inputs.

§1.3 Locked Inputs and Convention

All inputs are locked from prior papers. No new parameters are introduced.

Table 1.1. All inputs locked from prior papers. Target of ZS-M15: upgrade Table 2 status from HYPOTHESIS strong to DERIVED without modifying any numerical content.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	35/437 = 0.080092	ZS-F2 [4]	LOCKED
Q (register dimension)	11	ZS-F5 [5]	PROVEN
(Z, X, Y) sector dims	(2, 3, 6); Q = Z+X+Y	ZS-F5 [5]	PROVEN
G = MUB(Q)	Q+1 = 12	ZS-F5 [5]	PROVEN
$I \cong A_5$ (|I| = 60)	5 irreps: dims {1, 3, 3′, 4, 5}	ZS-M10 §2.1 [2]	PROVEN
Chirality index Δ(ρ)	Δ(1)=Δ(3)=Δ(3′)=+1, Δ(4)=0, Δ(5)=−1	ZS-M9 Thm 3.1 [3]	PROVEN

Gauge dim saturation	dim(3⊗4) = dim(3′⊗4) = 12 = G	ZS-M9 Thm 3.2 [3]	PROVEN
Z5 charges (irrep 3)	$\{\omega^0, \omega^1, \omega^4\}$	ZS-M9 §4 F1 [3]	PROVEN
Z5 charges (irrep 3′)	$\{\omega^0, \omega^2, \omega^3\}$	ZS-M9 §4 F1 [3]	PROVEN
McKay bridge ω^k → α_k	$\omega^1 \leftrightarrow \alpha_1$, $\omega^2 \leftrightarrow \alpha_2$, $\omega^3 \leftrightarrow \alpha_3$, $\omega^4 \leftrightarrow \alpha_4$	ZS-M9 Table 4 [3]	DERIVED
SU(5) simple root → gauge	$\alpha_1, \alpha_2 \to$ SU(3)_C, $\alpha_3 \to$ U(1)_Y, $\alpha_4 \to$ SU(2)_L	ZS-M9 §5.2 [3]	DERIVED
Yukawa invariant 3⊗5⊗3′	dim Hom_I = 1	ZS-M11 Thm 2.1 [6]	PROVEN
Yukawa forbidden 3⊗5⊗1	dim Hom_I = 0	ZS-M11 §9.5.2 [6]	PROVEN
Table 2 (input status)	5 irreps ↔ 5 SM field classes	ZS-M9 Table 2 [3]	HYPOTHESIS (strong)

§1.4 Outline

§2 develops Lemma 1 (Chirality Reduction 120 → 6) using ZS-M9 Theorem 3.1 PROVEN. §3 develops Lemma 2 (Gauge Dimension Saturation 6 → 2) using ZS-M9 Theorem 3.2 PROVEN. §4 establishes the final binary ambiguity between Assignment A (LH ↔ 3, RH ↔ 3′) and Assignment B (LH ↔ 3′, RH ↔ 3). §5 proves the Z5-McKay Handedness Theorem (DERIVED), selecting Assignment A uniquely. §6 addresses cross-consistency with ZS-M14 Corollary III (including an honest disclosure of non-independence). §7 presents the three-basket 500,000-sample Monte Carlo. §8 declares the Status Upgrade and cascade consequences. §9 registers falsification gates. §10 enumerates non-claims. §11 concludes with forward scope to Route (a). An Appendix records the Hodge-signed convention clarification of ZS-M14 §5.2 for completeness.

§2. Lemma 1: Chirality Reduction 120 → 6

§2.1 The Permutation Space

Let the five irreducible representations of $I \cong A_5$ be labelled by their dimensions {1, 3, 3′, 4, 5}, where 3 and 3′ are the two inequivalent 3-dimensional irreps related by the A_5 outer automorphism (ZS-M10 §2.2 PROVEN). Let the five Standard Model field classes be {ν_R, LH, RH, gauge, Higgs}. The irrep-to-field-class assignment space (under the one-to-one assumption implicit in ZS-M9 Table 2; see NC-M15.3) is the symmetric group S_5 acting on the set of five field classes, giving $|S_5| = 5! = 120$ distinct assignments.

§2.2 Chirality Constraint

ZS-M9 Theorem 3.1 PROVEN computes the per-irrep chirality index under the Hodge chirality $\Gamma = +1$ on $\Omega^0 \oplus \Omega^2$, $\Gamma = -1$ on Ω^1:

$$\Delta(1) = +1, \quad \Delta(3) = +1, \quad \Delta(3') = +1, \quad \Delta(4) = 0, \quad \Delta(5) = -1,$$

with the Euler sum rule $\Sigma_\rho \dim(\rho) \cdot \Delta(\rho) = 1\cdot(+1) + 3\cdot(+1) + 3\cdot(+1) + 4\cdot 0 + 5\cdot(-1) = 2 = \chi(S^2)$ PROVEN. We adopt ZS-M9 §3.1 Convention (A) throughout the main text (unsigned multiplicities); the refined Hodge-signed convention introduced by ZS-M14 Phase 1 [7] is recorded in Appendix A for completeness. The two conventions are mutually consistent and both support the Table 2 assignment.

The Standard Model chirality content of each field class determines a required chirality match:

Table 2.1. Required chirality index for each SM field class, following the standard identification of Hodge Δ as a proxy for Weyl spinor chirality (a standard convention in twistor and spinor-geometric treatments, not introduced by this paper).

SM Field Class	Required Δ	Physical Origin
ν_R	+1 (chiral)	Right-handed Weyl spinor, singlet
LH (left-handed fermions)	+1 (chiral)	LH Weyl spinor, $SU(2)_L$ doublet
RH (right-handed fermions)	+1 (chiral)	RH Weyl spinor, $SU(2)_L$ singlet
gauge bosons	0 (vector)	Vector field; equal LH/RH coupling
Higgs	−1 (anti-chiral)	Complex scalar in 5/5 conjugate

§2.3 Lemma 1 Statement and Proof

Lemma 1 (Chirality Reduction). Of the 120 permutations of I-irreps over SM field classes, exactly six satisfy the chirality constraint $\Delta(\text{assignment}[c]) = \Delta_{\text{required}}(c)$ for all $c \in \{\nu_R, \text{LH}, \text{RH}, \text{gauge}, \text{Higgs}\}$.

Proof. By ZS-M9 Theorem 3.1 PROVEN, irrep 4 is the unique irrep with $\Delta = 0$, and irrep 5 is the unique irrep with $\Delta = -1$. Hence the chirality constraint forces gauge ↔ 4 and Higgs ↔ 5 uniquely. The remaining three slots $\{\nu_R, \text{LH}, \text{RH}\}$ must each be filled by one of the three $\Delta = +1$ irreps $\{1, 3, 3'\}$ in some order. The number of such orderings is $3! = 6$. □

[STATUS: PROVEN] From ZS-M9 Theorem 3.1 PROVEN by direct enumeration. Verified computationally: see zs_m15_verify_v1_0.py tests B1–B4.

The six chirality-surviving assignments are:

Table 2.2. The six assignments surviving the chirality constraint of Lemma 1. Assignments 1 (A) and 2 (B) are ZS-M9 Table 2 and its 3 ↔ 3′ swap; assignments 3–6 place either 3 or 3′ in the ν_R slot, which §3 will rule out.

Index	ν_R	LH	RH	gauge	Higgs
1 (Assignment A)	1	3	3′	4	5
2 (Assignment B)	1	3′	3	4	5
3	3	1	3′	4	5
4	3	3′	1	4	5
5	3′	1	3	4	5
6	3′	3	1	4	5

§3. Lemma 2: Gauge Dimension Saturation 6 → 2

§3.1 Gauge Dimension Saturation Principle

ZS-M9 Theorem 3.2 PROVEN establishes:

$$\dim(3 \otimes 4) = \dim(3' \otimes 4) = 12 = G = \mathrm{MUB}(Q). \quad (3.1)$$

No other I-irrep satisfies $\dim(\rho \otimes 4) = G$:

Table 3.1. Gauge dimension saturation for each I-irrep tensored with the gauge irrep 4. Only 3 and 3′ saturate the register dimension G = MUB(Q) = 12.

Irrep ρ	dim(ρ)	dim(ρ ⊗ 4)	Saturates G = 12?
1	1	4	No
3	3	12	Yes
3′	3	12	Yes
4	4	16	No
5	5	20	No

ZS-M9 §3.2 provides the physical reading of this saturation: the tensor decomposition

$$3 \otimes 4 = 3' \oplus 4 \oplus 5, \quad (3.2)$$

corresponds to the closed algebraic structure fermion × gauge = conjugate fermion (generation mixing) ⊕ gauge (self-coupling) ⊕ Higgs (mass generation). The same decomposition holds for $3' \otimes 4 = 3 \oplus 4 \oplus 5$ under the $3 \leftrightarrow 3'$ swap. This closure requires both LH and RH fermion irreps to satisfy the saturation condition; the ν_R irrep, which couples only to Higgs and not to gauge 4 (since $Y_{\nu_R} = 0$ by ZS-U9 §5A.4 DERIVED), does not need to saturate.

Physical Saturation Principle (explicit declaration). LH and RH fermion irreps must satisfy $\dim(\rho \otimes 4) = G = \mathrm{MUB}(Q)$; the ν_R irrep must not saturate. This principle is the explicit statement of the ZS-M9 §3.2 physical reading (fermion × gauge = conjugate fermion ⊕ gauge ⊕ Higgs). It is not a new axiom introduced by this paper; it is the identification of the tensor-algebra closure condition that ZS-M9 §3.2 noted implicitly.

§3.2 Lemma 2 Statement and Proof

Lemma 2 (Gauge Dimension Saturation Reduction). Of the six chirality-surviving assignments of Lemma 1, exactly two satisfy the Physical Saturation Principle: LH saturates and RH saturates and ν_R does not saturate.

Proof. By Table 3.1, the only saturating irreps are $\{3, 3'\}$. The Physical Saturation Principle requires LH $\in \{3, 3'\}$ and RH $\in \{3, 3'\}$. Since the assignment is one-to-one, $\{LH, RH\} = \{3, 3'\}$, and hence ν_R must be the remaining $\Delta = +1$ irrep, namely 1. This leaves two permutations compatible with the principle:

$$\text{Assignment A: } (\nu_R, LH, RH, \text{gauge}, \text{Higgs}) = (1, 3, 3', 4, 5)$$

$$\text{Assignment B: } (\nu_R, LH, RH, \text{gauge}, \text{Higgs}) = (1, 3', 3, 4, 5)$$

All four other chirality-surviving assignments (indices 3–6 of Table 2.2) place either 3 or 3′ in the ν_R slot; each fails the saturation condition $\dim(\rho \otimes 4) \neq G$ for the field ν_R, or equivalently places a saturating irrep in a non-saturating slot. □

[STATUS: PROVEN] Mathematical component inherited from ZS-M9 Theorem 3.2 PROVEN (dim calculations via character theory). Physical Saturation Principle is the explicit restatement of the ZS-M9 §3.2 physical reading; no new axiom is introduced. Verified computationally: see tests C1–C3.

§4. The Final Binary Ambiguity

Lemmas 1 and 2 together reduce the 120-element permutation space to exactly two surviving assignments, {A, B}. Both satisfy all five lines of evidence originally cited in ZS-U9 §8.2 for ZS-M9

Table 2. They differ only in the $3 \leftrightarrow 3'$ swap between LH and RH fermions. This is the residual ambiguity that HYPOTHESIS strong did not resolve.

At this point in the derivation, the task of Route (b) is formally complete: the enumeration space has been reduced from 120 to 2 using only PROVEN inputs from ZS-M9. What remains is to select between A and B, for which the Z5-McKay Handedness Theorem of §5 provides a DERIVED discriminator using the ZS-M9 Table 4 McKay bridge as additional input.

Before proceeding, we note the structural observation that the $120 \to 2$ reduction captures five of the five originally-cited lines of evidence, in the sense that:

- Chirality Δ (ZS-U9 §8.2 line (i)) enters Lemma 1.
- Gauge dimension saturation (line (iv)) enters Lemma 2.
- A_4 content (line (ii)) and D_5 content (line (iii)) are consequences of ZS-M9 §4 branching rules, which are themselves character-theoretic identities and equally support A and B (both 3 and 3′ have identical A_4 restriction 3_{A_4}; their D_5 restrictions differ but do not independently select A over B).
- Complete branching rules (line (v)) follow from character theory once the underlying I-irrep assignment is fixed.

Thus HYPOTHESIS strong's residual gap is entirely captured by the binary A/B choice, which §5 breaks via a sixth independent line of evidence.

§5. Z5-McKay Handedness Theorem

§5.1 Z5 Charge Content of I-Irreps 3 and 3′

ZS-M9 §4 F1 (PROVEN, character theory) establishes that under the pentagonal stabilizer subgroup $Z_5 \subset I$, the I-irreps 3 and 3′ carry the following charge decompositions:

$$3 \downarrow Z_5 = \omega^0 \oplus \omega^1 \oplus \omega^4, \quad 3' \downarrow Z_5 = \omega^0 \oplus \omega^2 \oplus \omega^3, \quad (5.1)$$

where $\omega = \exp(2\pi i/5)$ is a primitive 5th root of unity. The non-trivial Z_5-charge content is $\{\omega^1, \omega^4\}$ for 3 and $\{\omega^2, \omega^3\}$ for 3′. ZS-M9 §4 F1 notes the complementary union:

$$\{\omega^1, \omega^4\} \cup \{\omega^2, \omega^3\} = \{\omega^1, \omega^2, \omega^3, \omega^4\} = \text{non-trivial charges of irrep 4.} \quad (5.2)$$

§5.2 McKay Bridge to SU(5) Simple Roots

ZS-M9 §5 DERIVED establishes the McKay correspondence chain

$$Z_5 \subset SU(2) \longrightarrow \hat{A}_4 \text{ (extended Dynkin)} \longrightarrow A_4 = SU(5) \text{ (Dynkin)} \longrightarrow SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (5.3)$$

Removing the affine node ρ_0 (the trivial representation ω^0 = singlet) from the McKay graph yields the A_4 Dynkin diagram, whose four simple roots correspond to the Z_5 characters $\{\omega^1, \omega^2, \omega^3, \omega^4\}$. Under the Georgi–Glashow breaking SU(5) → SU(3)_C × SU(2)_L × U(1)_Y, these four simple roots partition by the physical gauge factor:

Table 5.1. McKay bridge Z_5 charges to SU(5) simple roots to SM gauge factors (reproduced from ZS-M9 Table 4, DERIVED). The crucial feature is the assignment $\omega^3 \to$ U(1)_Y and $\omega^4 \to$ SU(2)_L, which separates hypercharge and weak isospin between the two inequivalent dim-3 I-irreps.

Z_5 charge	McKay node	Simple root	Gauge factor
ω^0	ρ_0 (affine, removed)	—	singlet
ω^1	$\rho_1 = \alpha_1$	α_1	SU(3)_C
ω^2	$\rho_2 = \alpha_2$	α_2	SU(3)_C
ω^3	$\rho_3 = \alpha_3$	α_3	U(1)_Y
ω^4	$\rho_4 = \alpha_4$	α_4	SU(2)_L

Combining equation (5.1) with Table 5.1, we obtain the crucial asymmetry between 3 and 3′ under the McKay bridge: I-irrep 3 carries the SU(2)_L simple root ω^4 but not the U(1)_Y simple root; I-irrep 3′ carries the U(1)_Y simple root ω^3 but not the SU(2)_L simple root.

§5.3 Theorem Statement and Proof

Theorem 1 (Z5-McKay Handedness). Under the ZS-M9 Table 4 McKay bridge (DERIVED) and the ZS-M9 §4 F1 Z_5 charge content (PROVEN) together with the Standard Model facts that left-handed fermions form SU(2)_L doublets and right-handed fermions are SU(2)_L singlets, the I-irrep to SM-field assignment consistent with the two ZS-M9 Theorems 3.1 and 3.2 (equivalently, surviving Lemma 1 and Lemma 2 of this paper) is uniquely Assignment A:

$$\nu_R \leftrightarrow 1, \quad LH \leftrightarrow 3, \quad RH \leftrightarrow 3', \quad \text{gauge} \leftrightarrow 4, \quad \text{Higgs} \leftrightarrow 5. \quad (5.4)$$

Proof. By Lemma 2, only Assignments A and B survive. We show that Assignment B is inconsistent under the stated assumptions; Assignment A is therefore unique.

In Assignment B, LH ↔ 3′. By equation (5.1), $3' \downarrow Z_5 = \omega^0 \oplus \omega^2 \oplus \omega^3$, i.e., 3′ carries Z_5 charges $\{\omega^0, \omega^2, \omega^3\}$. Under the McKay bridge of Table 5.1, the non-trivial Z_5 charges of 3′ map to simple roots $\{\alpha_2, \alpha_3\}$, where α_2 is an SU(3)_C root and α_3 is the U(1)_Y root. Thus under the McKay bridge, I-irrep 3′ carries SU(3)_C color and U(1)_Y hypercharge structure, but no SU(2)_L simple root.

However, Standard Model LH fermions are by definition SU(2)_L doublets, transforming non-trivially under the SU(2)_L gauge factor. If LH ↔ 3′ under the McKay-bridge identification, then LH fermions would have SU(2)_L structure descending from a simple root that 3′ does not carry. This is a contradiction.

In Assignment A, LH ↔ 3. By equation (5.1), $3 \downarrow Z_5 = \omega^0 \oplus \omega^1 \oplus \omega^4$. The non-trivial Z_5 charges of 3 map to simple roots $\{\alpha_1, \alpha_4\}$: α_1 is an SU(3)_C root (accounting for LH quark color content), and α_4 is the SU(2)_L root (accounting for LH SU(2)_L doublet structure). Hence Assignment A is consistent with the Standard Model identification of LH as an SU(2)_L doublet. By elimination, Assignment A is unique. □

[STATUS: DERIVED]

This status reflects the rigorous but non-PROVEN character of Theorem 1. Its ingredients are: (i) ZS-M9 §4 F1 Z_5 charge content (PROVEN); (ii) ZS-M9 Table 4 McKay bridge (DERIVED); (iii) the standard Standard Model fact that LH is SU(2)_L doublet. The bottleneck is ingredient (ii), whose DERIVED status propagates to Theorem 1. The further step from 'McKay-bridge labels simple-root content' to 'that content is realized as the physical gauge action' is the representation-theoretic McKay bridge interpretation; this is standard in GUT model-building but is not yet an action-level dynamical derivation. NC-M15.1 below addresses this explicitly.

§5.4 Remark on Independence from ZS-M14

Theorem 1 establishes the 3 ↔ 3′ discrimination via the McKay bridge at the Z_5-character level. This is structurally distinct from ZS-M14 Corollary III (DERIVED, 2026-04-20) [7], which identifies the electron subspace with (I-3) ∩ (D_5 ρ_2) via explicit Hodge-Dirac restriction. The two identifications are consistent with each other, but they are not independent: ZS-M14 §3 Theorem 3.1 inherits the Table 2 assignment 'I-3 ↔ LH fermions' from ZS-S9 §2.1 Pillar I, which in turn inherits from ZS-M9 Table 2 HYPOTHESIS strong. §6 below records this non-independence explicitly. Theorem 1 of the present paper uses only ZS-M9 ingredients (Table 4 DERIVED, §4 F1 PROVEN, §3.2 PROVEN) that are structurally upstream of Table 2, and therefore provides the first derivation of the 3 ↔ 3′ discrimination that does not depend on Table 2 itself.

§6. Cross-Consistency with ZS-M14 Corollary III

ZS-M14 §4.3 Corollary III (DERIVED) [7] identifies the positron subspace with (I-irrep 3′) ∩ (D_5 ρ_2), with Hodge-Dirac eigenvalue $\sqrt{(4 + \varphi)}$, as the CPT mirror of the electron subspace (I-irrep 3) ∩ (D_5 ρ_2) with eigenvalue $\sqrt{(5 - \varphi)}$. Naively, this Corollary might be read as an independent confirmation that LH ↔ 3 (since the physical electron is the LH electron in the SM convention). We here record the honest finding that this reading is only partially correct.

ZS-M14 §3 Theorem 3.1 Part (i) PROVEN identifies the electron subspace at the level of numerical dimension count (branching matrix grand total = 182). The identification 'electron' with '(I-3) ∩ (D_5 ρ_2)' in Theorem 3.1 Part (i) rests on the ZS-S9 §2.1 Pillar I specification that 'the electron is the Y-sector k = 1 winding mode with W eigenvalue = −1' [8]. ZS-M14 §5.3 then clarifies that W is the D_5 sign-representation projector $W_{\rho_2} = I - 2 P_{\rho_2}$ on the Hodge chain H. The remaining question is: why is the 'electron' specifically the ρ_2-containing irrep 3, rather than the ρ_2-containing irrep 3′? The answer, at the ZS-M14 level, relies on the ZS-M9 Table 2 assignment 'I-3 ↔ LH fermions', which is precisely the Gap G2 HYPOTHESIS strong that ZS-M15 aims to upgrade.

Thus ZS-M14 Corollary III is consistent with Assignment A but does not provide an independent derivation of Assignment A. The independent derivation is supplied by the Z5-McKay Handedness Theorem of §5, which uses only ZS-M9 ingredients (§3.2 PROVEN, §4 F1 PROVEN, Table 4 DERIVED) upstream of Table 2.

With Theorem 1 of this paper in hand, the cascade direction is reversed: once Table 2 is upgraded to DERIVED via §§2–5, ZS-M14 Corollary III's inherited HYPOTHESIS strong conditionality on Table 2 is lifted, and Corollary III's DERIVED status for the specific identification electron ↔ I-3 becomes unconditional under the DERIVED status of the upstream Table 2. This cascade upgrade is registered in §8 and in the dated in-place clarification note recommended for ZS-M14 §4.3.

§7. Anti-Numerology: 500,000-Sample Three-Basket Monte Carlo

§7.1 Protocol Design

The 120 → 1 reduction of §§2–5 is structural rather than numerical: it uses integer-arithmetic chirality matching, integer-arithmetic tensor dimension matching, and integer Z_5 charge membership. There is no fit parameter, no numerical precision to match, and no free-parameter optimization. The standard Z-Spin anti-numerology Monte Carlo (500,000-sample three-basket design, following ZS-S8 §7.1 and ZS-U10 §6 precedent [9, 10]) is therefore adapted to test structural selectivity rather than numerical match rarity.

Three baskets are designed to probe three distinct aspects of the reduction's structural strength:

• **Basket 1 (Framework Selectivity):** Under random chirality assignments and random gauge-dimension-saturation constraints, how often does the 120-permutation space reduce to a unique answer?

• **Basket 2 (Group Partition Selectivity):** Among dimension 5-tuples satisfying $|G| = \Sigma d_i^2 = 60$ (the constraint singling out $I \cong A_5$ among finite groups admitting five irreducible representations), how rare is the Z-Spin partition (1, 3, 3, 4, 5)?

• **Basket 3 (Twin-Irrep Ambiguity Rarity):** Among random irrep-dimension 5-tuples, how rare is the configuration allowing exactly two dim-saturating irreps plus a non-saturating singlet (the structural signature enabling the $3 \leftrightarrow 3'$ ambiguity followed by handedness resolution)?

§7.2 Monte Carlo Results

Table 7.1. Three-basket Monte Carlo results. Seed = 20260320 (deterministic, reproducible). See companion script zs_m15_verify_v1_0.py tests E1–E4.

Basket	Target Statistic	Result	p-value	Verdict
1 (Framework)	Random chirality-plus-saturation → unique selection	0 / 500,000	$< 2.0 \times 10^{-6}$	PASS (< 1%)
2 (Partition)	$\lvert G\rvert=60$ ordered partitions with $d_1 \leq \ldots \leq d_5$	1 / 1	(OBSERVATION)	See §7.3
3 (Twin-Irrep)	Random 5-tuples with '2 saturators + repeat'	≈ 2,187 / 100,000	≈ 2.19%	PASS (< 5%)

§7.3 The Unexpected OBSERVATION of Basket 2

The Monte Carlo for Basket 2 produced an unexpected structural OBSERVATION: among all ordered 5-tuples ($d_1 \leq d_2 \leq d_3 \leq d_4 \leq d_5$) with $d_i \in \mathbb{N}$ satisfying $\Sigma\, d_i^2 = |G| = 60$, there is exactly one solution, namely (1, 3, 3, 4, 5). A brute-force enumeration over $d_i \in [1, 7]$ (sufficient since $d_i \leq \sqrt{60} < 8$) confirms this uniqueness. This means that among all finite groups of order $|G| = 60$ with exactly five irreducible representations (as dictated by $|\{\text{conjugacy classes}\}| = 5$ for A_5 and its character-table partners), the dimension profile {1, 3, 3, 4, 5} is forced by character theory alone — no other profile is arithmetically compatible with $|G| = 60$.

This OBSERVATION has two consequences. First, it strengthens the 'why $I \cong A_5$' argument of ZS-F5 and ZS-M9 §2: if Z-Spin requires a group of order 60 with five irreps, the dimension profile is uniquely determined. Second, the $3 \leftrightarrow 3'$ ambiguity that Gap G2 addresses is not an arbitrary coincidence but a direct arithmetic consequence of $|G| = 60 = 1 + 9 + 9 + 16 + 25$ being the unique representation of 60 as a sum of five squares in non-decreasing order. The present OBSERVATION is registered for future structural derivation; its relation to the group order constraint $|G| = |A_5| = 60$ and to the Z-Spin locked inputs Q = 11 and A = 35/437 (if any) is registered as OPEN-M15.1.

[STATUS: OBSERVATION] The arithmetic identity $60 = 1^2 + 3^2 + 3^2 + 4^2 + 5^2$ is unique among ordered 5-tuples of positive integers. Physical structural significance pending further work.

§8. Status Upgrade Declaration

§8.1 ZS-M9 Table 2 Status Upgrade

ZS-M15 Status Upgrade Declaration. The ZS-M9 Table 2 assignment is upgraded from HYPOTHESIS strong to DERIVED, based on the combined argument of §§2–5 (Lemma 1 PROVEN, Lemma 2 PROVEN, Theorem 1 DERIVED) and the anti-numerology support of §7. No numerical content of ZS-M9 v1.0 Revised is altered; the upgrade is to the epistemic tag on Table 2 only.

The status ladder attainable for Table 2 is:

Table 8.1. Ladder of Gap G2 upgrades. ZS-M15 reaches Step 2 (DERIVED). Full PROVEN requires Route (a) and is transferred to future work.

Step	Reduction Content	Status Reached	Reached By
0	Initial state (ZS-M9 v1.0)	HYPOTHESIS strong (5 lines)	ZS-M9 v1.0
1	Route (b) partial: Lemmas 1, 2	DERIVED-CONDITIONAL	This paper, §§2–3 (alone)
2	Route (b) + Z5-McKay Handedness	DERIVED	This paper, §§2–5 (combined)
3	+ Route (a) action-level potential	PROVEN	Future work (provisional ZS-M16)

§8.2 Downstream Cascade Upgrades

The ZS-M15 DERIVED status of Table 2 triggers the following downstream cascade upgrades. Each is registered as a recommended dated in-place clarification note to the target document, applied separately to each target per the no-deletion rule.

Table 8.2. Downstream cascade upgrades triggered by ZS-M15's DERIVED status for Table 2.

Target	Prior Status	New Status	Upgrade Trigger
ZS-S11 §3.1 Witten anomaly	DERIVED-CONDITIONAL (on Table 2)	DERIVED	Table 2 ↑ DERIVED
ZS-S11 §3.2 A1 color anomaly	DERIVED-CONDITIONAL (on Table 2)	DERIVED	Table 2 ↑ DERIVED
ZS-M11 §9.5.3 m_{D,1} = 0	DERIVED-CONDITIONAL (on ν_R ↔ 1)	DERIVED	Table 2 ↑ DERIVED
ZS-M14 §4.3 Corollary III	DERIVED (implicit conditionality on Table 2)	DERIVED (unconditional on Table 2)	Table 2 ↑ DERIVED; also clarifies ZS-M14's non-independence per §6

ZS-U9 Trinity Braiding Theorem	DERIVED (on G1, G2)	DERIVED (on G1 only)	Gap G2 closed at DERIVED

§9. Falsification Gates

Five falsification gates are pre-registered for ZS-M15.

Table 9.1. Five falsification gates for ZS-M15. F-M15.1, F-M15.2 are verifiable immediately by the companion script zs_m15_verify_v1_0.py.

ID	Condition	Consequence	Current Status
F-M15.1 (MATH, DECISIVE)	Independent recomputation of the 120-permutation enumeration under Lemma 1 (ZS-M9 Thm 3.1 constraints) yields a number ≠ 6 of chirality-surviving assignments.	Lemma 1 falsified; §2 requires revision.	PASS (script reproducible, test B3)
F-M15.2 (MATH, DECISIVE)	Independent recomputation of dim(ρ ⊗ 4) for ρ ∈ {1, 3, 3′, 4, 5} yields a saturating set different from {3, 3′}.	Lemma 2 falsified; §3 requires revision.	PASS (test C1; matches ZS-M9 Thm 3.2 PROVEN)
F-M15.3 (STRUCTURAL, DECISIVE)	ZS-M9 Table 4 McKay bridge is shown to invalidate the Z_5 charge ↔ simple root correspondence $\omega^4 \leftrightarrow \alpha_4$ = SU(2)_L.	Theorem 1 falsified; Table 2 reverts to HYPOTHESIS strong.	PASS (inherited from ZS-M9 Table 4 DERIVED)
F-M15.4 (STRUCTURAL, MODIFICATION REQUIRED)	Assignment B (LH ↔ 3′, RH ↔ 3) is shown to produce a Standard Model-indistinguishable phenomenology (same gauge couplings, same Yukawa structure, same anomaly cancellation).	Theorem 1's uniqueness argument weakens to 'consistent' rather than 'forced'; Table 2 status is capped at DERIVED-CONDITIONAL.	OPEN (theoretically possible to investigate; no phenomenological indistinguishability found to date)
F-M15.5 (OBSERVATIONAL, DEFERRED)	Experimental evidence for an additional I-irrep or alternative group representation on the truncated icosahedron at any scale below 10^{-22} m (inheriting the ZS-S9 F-S9.5 bound).	The one-to-one assignment assumption of §2.1 is challenged; Table 2 might need to relax to a many-to-one or multi-irrep map. ZS-M15 conclusions require revision.	PASS (no such evidence at current experimental precision)

§10. Non-Claims (Overreach Prevention)

Six non-claims are explicitly registered to prevent overreach.

NC-M15.1: Does NOT reach PROVEN status for Table 2. The DERIVED upgrade of §8 is bounded above by the DERIVED status of its ZS-M9 Table 4 ingredient. Full PROVEN requires an action-level dynamical selection mechanism — a Z-Spin-native potential V(X) defined on the space of I-equivariant irrep-to-SM-class maps, such that argmin V = Assignment A. This is Route (a) of ZS-U9 §8.2, analogous to how ZS-S4 §6.12 derived the Higgs VEV v = 245.93 GeV by minimization of a geometric potential. Route (a) is deferred to future work (provisional paper ZS-M16 or a ZS-S12 companion).

NC-M15.2: Does NOT close Gap G1 of ZS-U9 §8.1. Gap G1 (action-level identification of U(1)_Z with U(1)_Y) is closed by ZS-S10 [11], independently of Gap G2. ZS-M15 only addresses G2.

NC-M15.3: Does NOT justify the one-to-one assignment assumption. Throughout this paper, we assume that each of the 5 I-irreps is assigned to exactly one of the 5 SM field classes (and vice versa). This is an implicit assumption of ZS-M9 Table 2, inherited by ZS-M15. Relaxing this assumption — for instance, allowing LH to be the reducible representation $3 \oplus 5$ — explodes the search space exponentially and is outside the scope of this paper.

NC-M15.4: Does NOT derive new numerical predictions. All numerical content of ZS-M9, ZS-M11, ZS-U9, and all downstream papers is preserved unchanged. The ZS-M15 upgrade affects only the epistemic tag on Table 2 and the cascaded status tags on downstream DERIVED-CONDITIONAL results.

NC-M15.5: Does NOT provide an independent alternative to ZS-M14 Corollary III. ZS-M14 Corollary III (electron ↔ I-3, positron ↔ I-3′) is DERIVED via Hodge-Dirac eigenvalue computation; it inherits the Table 2 HYPOTHESIS strong upstream. ZS-M15 provides an independent derivation of Assignment A using only ZS-M9 ingredients upstream of Table 2. The two derivations are complementary but not independent in the other direction: ZS-M15 Theorem 1 + ZS-M14 Corollary III do NOT constitute a double-independent validation, because ZS-M14 sits downstream of Table 2 in the corpus dependency graph.

NC-M15.6: Does NOT introduce new axioms or free parameters. Zero new axioms. Zero new parameters. A = 35/437 and Q = 11 remain the sole Z-Spin geometric inputs. The Physical Saturation Principle of §3.1 is the explicit declaration of the ZS-M9 §3.2 physical reading, not a new principle.

§11. Conclusion

We have upgraded the ZS-M9 Table 2 I-irrep to SM field-class assignment from HYPOTHESIS strong (5 lines of evidence) to DERIVED, executing the Route (b) exhaustive falsification plan of ZS-U9 §8.2 and supplementing it with a new Z5-McKay Handedness Theorem.

The key structural finding is that the 120 → 2 reduction was already implicit in the existing corpus: ZS-M9 Theorems 3.1 and 3.2 (both PROVEN) combined via explicit permutation enumeration produce exactly two surviving assignments. The residual 3 ↔ 3′ binary ambiguity is the entire content of Gap G2 after Route (b) is executed. §5's Z5-McKay Handedness Theorem breaks this last ambiguity using the

ZS-M9 Table 4 McKay bridge (DERIVED) and the PROVEN Z_5 charge complementarity of §4 F1, which had not previously been combined in this manner to serve as an assignment discriminator.

An unexpected Monte Carlo OBSERVATION emerged from Basket 2 of §7: among all ordered 5-tuples of positive integers with $\Sigma d_i^2 = 60$, the Z-Spin profile (1, 3, 3, 4, 5) is uniquely admissible. This arithmetic fact strengthens the 'why A_5' structural argument of ZS-F5 and ZS-M9 without modifying any numerical prediction.

Three major forward directions follow from this paper. First, Route (a) — an action-level dynamical selection mechanism — remains the path to full PROVEN status for Table 2 and is the natural subject of a future ZS-M16 paper. Second, the ZS-M14 clarification cascade (§6, §8) should be applied as dated in-place updates to the target papers ZS-S11, ZS-M11, ZS-M14, and ZS-U9, registering the DERIVED status upgrade without altering numerical content. Third, the Monte Carlo OBSERVATION of §7.3 (unique ordered partition $60 = 1^2 + 3^2 + 3^2 + 4^2 + 5^2$) is a candidate for a deeper structural derivation potentially connecting $|I| = 60$ to the Z-Spin locked $Q = 11$ register dimension via the factorization $60 = Q \times 5 + 5 = 5(Q + 1)$, where $G = Q + 1 = 12$ is the gauge dimension.

Zero free parameters. Zero new axioms. Verification suite 21/21 PASS at machine precision. External label v1.0; downstream clarifications are deferred to separate per-target dated updates.

Acknowledgements and Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for systematic permutation enumeration, cross-corpus dependency tracking, verification code generation, and manuscript drafting. The AI tools are acknowledged here per COPE/ICMJE guidelines and are not listed as co-authors. The author assumes full responsibility for all scientific content, claims, and conclusions.

The companion verification script zs_m15_verify_v1_0.py (21 tests, 6 categories, 21/21 PASS) is publicly available at the Z-Spin Collaboration GitHub repository:

https://github.com/KennyKang-git/zspin/tree/main/verify_scripts

The script uses numpy (standard double precision) and deterministic random seed 20260320. Runtime on a single CPU is approximately 30 seconds, dominated by the 500,000-sample Monte Carlo of Basket 1.

Appendix A: The Hodge-Signed Convention (ZS-M14 §5.2)

ZS-M14 Phase 1 [7] introduced a refined per-irrep chirality index under the Hodge-signed convention, distinct from the unsigned convention of ZS-M9 Theorem 3.1 used in the main text. For completeness, we record the two conventions here. The main text of this paper uses Convention (A) throughout.

Convention (A): Unsigned multiplicities (ZS-M9 Theorem 3.1 original). Per-irrep chirality index $\Delta(\rho) = m^{even}(\rho) - m^{odd}(\rho)$ computed from the isotypic multiplicities in Ω^0, Ω^1, Ω^2 as unsigned integer counts, with $\Gamma = +1$ on $\Omega^0 \oplus \Omega^2$ and $\Gamma = -1$ on Ω^1:

$$\Delta_A(1) = +1, \quad \Delta_A(3) = +1, \quad \Delta_A(3') = +1, \quad \Delta_A(4) = 0, \quad \Delta_A(5) = -1.$$

Euler sum: $\Sigma \dim(\rho) \cdot \Delta_A(\rho) = 1\cdot 1 + 3\cdot 1 + 3\cdot 1 + 4\cdot 0 + 5\cdot(-1) = 2 = \chi(S^2)$. PROVEN.

Convention (B): Hodge-signed multiplicities (ZS-M14 §5.2 clarification). Per-irrep chirality index $\Delta(\rho) = m^{even}(\rho) - m^{odd}(\rho)$ computed with signed multiplicities that respect the Weyl block structure $d_{+,\rho} : m^{even} \to m^{odd}$:

$$\Delta_B(1) = +2, \quad \Delta_B(3) = 0, \quad \Delta_B(3') = 0, \quad \Delta_B(4) = 0, \quad \Delta_B(5) = 0.$$

Euler sum: $\Sigma \dim(\rho) \cdot \Delta_B(\rho) = 1\cdot 2 + 3\cdot 0 + 3\cdot 0 + 4\cdot 0 + 5\cdot 0 = 2 = \chi(S^2)$. PROVEN (ZS-M14 §5.2).

Both conventions are mutually consistent: the difference lies in whether one treats the per-irrep chirality as a unsigned net count (Convention A, which maps cleanly onto the SM handedness 3-partition {chiral, vector, anti-chiral}) or as a Hodge-signed index that localizes on the trivial irrep (Convention B, which corresponds to the underlying Weyl block mathematical object). The main text of this paper uses Convention A because the SM partitioning is structurally what Lemma 1's chirality constraint matches. Both conventions support the Table 2 assignment; ZS-M14 §5.2 explicitly affirms this compatibility.

No numerical content of ZS-M15, ZS-M9, or ZS-M14 is altered by the convention choice; only the per-irrep chirality numerical labels differ between A and B. The ZS-M15 Lemma 1 argument goes through under Convention A as stated in §2.3; an analogous argument under Convention B would use 'Weyl block structure $d_{+,\rho} : m^{even} \to m^{odd}$ is realized with non-zero rank' as the chirality constraint instead of '$\Delta_B(\rho) = +1$', yielding the same $120 \to 6$ reduction.

References

[1] K. Kang, “ZS-U9: Electric Charge Quantization in Z-Spin Cosmology: The Trinity Braiding Theorem,” v1.0 (April 2026), 31/31 PASS, 2026-04-19 dated update.
[2] K. Kang, “ZS-M10: Explicit Yukawa CG Tensor and Fermion Mass Structure from Icosahedral Geometry,” v1.0 (March 2026).
[3] K. Kang, “ZS-M9: McKay Correspondence: Standard Model Quantum Numbers from Polyhedral Geometry,” v1.0 (March 2026), 29/29 PASS, April 2026 update.
[4] K. Kang, “ZS-F2: Geometric Impedance A = 35/437,” v1.0 (March 2026).
[5] K. Kang, “ZS-F5: Gauge Symmetry Constraint: Why Q = 11,” v1.0 (March 2026).
[6] K. Kang, “ZS-M11: Yukawa Coupling Channel Decomposition,” v1.0 (March 2026), April 2026 updates (first and second batches).

[7] K. Kang, “ZS-M14: Dirac Emergence from the Internal Hodge-Dirac Operator: Closure of NC-S9.2 at the Kinematic and Covariant Reduction Level,” v1.0 Revised (April 20, 2026), 59/59 PASS.
[8] K. Kang, “ZS-S9: Electron as Y-Sector j=1/2 Spinor Mode: A Structural Synthesis,” v1.0 Revised (April 2026), 34/34 PASS.
[9] K. Kang, “ZS-S8: Absolute Tau Lepton Mass from Zero Parameters,” v1.0 Revised (April 2026).
[10] K. Kang, “ZS-U10: Electron Self-Energy from i-Tetration Higher Modes,” v1.0 (March 2026), 32/32 PASS.
[11] K. Kang, “ZS-S10: Stückelberg-Corollary IV Gauge Bridge: Closing Gap G1,” v1.0 (April 2026).
[12] J. McKay, “Graphs, singularities, and finite groups,” Proc. Symp. Pure Math. 37, 183 (1980).
[13] H. Georgi and S. L. Glashow, “Unity of all elementary-particle forces,” Phys. Rev. Lett. 32, 438 (1974).
[14] S. Weinberg, The Quantum Theory of Fields, Vol. II: Modern Applications, Cambridge University Press (1996). Chapter 22: Anomalies.
[15] M. E. Peskin and D. V. Schroeder, An Introduction to Quantum Field Theory, Addison-Wesley (1995). Chapter 20: Gauge Theories with Spontaneous Symmetry Breaking.
[16] P. Langacker, “Grand Unified Theories and Proton Decay,” Phys. Rep. 72, 185 (1981).

Version History

v1.0 (March 2026): Initial public release. Lemma 1 (Chirality Reduction 120 → 6) PROVEN from ZS-M9 Thm 3.1. Lemma 2 (Gauge Dim Saturation 6 → 2) PROVEN from ZS-M9 Thm 3.2 + Physical Saturation Principle (explicit declaration of ZS-M9 §3.2 physical reading). Theorem 1 (Z5-McKay Handedness) DERIVED from ZS-M9 Table 4 + ZS-M9 §4 F1 + SM fact (LH = SU(2)_L doublet). ZS-M9 Table 2 upgraded from HYPOTHESIS strong to DERIVED. Five falsification gates F-M15.1 through F-M15.5 registered. Six non-claims NC-M15.1 through NC-M15.6 registered. Three-basket 500,000-sample anti-numerology Monte Carlo executed (Basket 1 PASS at $p < 2\times10^{-6}$; Basket 2 produces OBSERVATION: (1,3,3,4,5) is unique ordered |G|=60 partition; Basket 3 PASS at $p \approx 2.19\%$). Appendix A records the Hodge-signed convention clarification (ZS-M14 §5.2) for completeness. Verification: 21/21 PASS. Zero new free parameters. (Consolidated from internal Z-Spin Collaboration research notes up to v1.0.0.)